

Finding Hamiltonian Cycle on Cartesian Product of a Hypotractable Graph and a Hamiltonian Graph

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Abstract

The interconnected network is usually represented by a graph where the vertices represent processors and the edges represent links between processors. A hypotractable graph G is a graph with no Hamiltonian path, and for any vertex $v \in V(G)$, $G - v$ has a Hamiltonian path. This paper investigates the Cartesian product of a hypotractable graph with a cycle C_n . It is shown that the Cartesian product of a hypotractable graph and a Hamiltonian graph is Hamiltonian.

Keywords: Hamiltonian, hypotractable, Cartesian product.

1 Introduction

An interconnected network for parallel and distributed system is usually represented by a graph $G = (V, E)$, where the vertices represent processors and the edges represent links between processors [3,4,9]. Cycle embedding problems in the interconnected networks has been studied for a long time. One of the important properties and major requirements is designing the topology of networks with Hamiltonian property [2]. A path which contains every vertex of G is called a Hamiltonian path of G . Similarly, a cycle that contains every vertex of G is called a Hamiltonian cycle of G . Such paths and cycles are described after Hamilton (1856) [1]. A graph is Hamiltonian if it contains a

Hamiltonian cycle. A graph is Hamiltonian connected if it contains a Hamiltonian path. The graph $G \times H$ is the Cartesian product of the graphs $G = (V_g, E_g)$ and $H = (V_h, E_h)$ with vertex set $V(G \times H) = \{(u, v) \mid \forall u \in V_g \text{ and } v \in V_h\}$ edge set $E(G \times H) = \{((u_1, v_1), (u_2, v_2))\}$ if $((u_1, u_2) \in E$ and $v_1 = v_2)$ or $(u_1 = u_2, \text{ and } (v_1, v_2) \in E_h)\}$ [5]. A hypotractable graph G is a graph which has no Hamiltonian path, but for any vertex $v \in V(G)$, $G - v$ has a Hamiltonian path. A hypo-Hamiltonian graph G is a graph which has no Hamiltonian cycle, but for any vertex $v \in V(G)$, $G - v$ has a Hamiltonian cycle. Carsten constructed the schema of Hypotractable graphs by four disjoint hypo-Hamiltonian graphs of degree 3 and proved that there exist infinitely many planar cubic Hypotractable graphs and many finite, non-planar, cubic hypotractable graphs [6-8]. In this paper, we show that $G \times C_n$ is Hamiltonian where G is a hypotractable graph. Furthermore, we also obtain the corollary that $G_1 \times G_2$ is a Hamiltonian graph which G_1 is a hypotractable graph and G_2 is a Hamiltonian graph.

2 Cartesian Product of Hypotractable Graph and Cycle

Thomassen proved that for $p = 34, 37, 39, 40$ and for all $p \geq 42$, there exists a hypotractable graph with p vertices [6]. The hypotractable graph of order 34 is then called *Thomassen Graph*.

Lemma 1 Let G be a hypotractable graph. Then every vertex v in G , $\deg_G v \geq 3$.

Proof. Suppose that G is a hypotractable graph with a vertex v such that $\deg_G v < 3$. If $\deg_G v = 1$, let u be the vertex adjacent to v . Then $G - u$ has an isolated vertex v , hence there is no Hamiltonian path in $G - u$ which contradict to G is hypotractable. If $\deg_G v = 2$, let u be a vertex adjacent to v . Then v must be either the starting or ending vertex of the Hamiltonian path in $G - u$. Then this path with edge vu forms a Hamiltonian path in G which contradict to G is hypotractable. ■

Theorem 1 Let G be a hypotractable graph of order k . Then the graph $G \times C_n$ for $n \geq 3$ admits a Hamiltonian cycle.

Proof. Let $G_1, G_2, \dots, G_n$ be n identical copies of graph G in $G \times C_n$. Let $P_1: v_1^1, v_2^1, \dots, v_{k-1}^1$; $P_2: v_1^2, v_2^2, \dots, v_{k-1}^2$; ...; $P_n: v_1^n, v_2^n, \dots, v_{k-1}^n$ be n identical Hamiltonian paths of $G_1 - v^1, G_2 - v^2, \dots, G_n - v^n$, respectively. Suppose v_u^t, v_i^t, v_j^t for $u < i < j$ be three vertices adjacent to v^t in G_t for $1 \leq t \leq n$, consider following cases:

Case 1.1. $n, u-1$ and $k-1-j$ are all even. Then consider a cycle from v_1^1 to v_i^1 , then v^1 and v_j^1 , back to v_{i+1}^1 , then jump to the corresponding vertex v_{i+1}^2 in G_2 , go through subpath of P_2 to v_{j+1}^2 , then go back and forth in G_1 and G_2 until v_{k-1}^2 , then go through all v_{k-1}^t for $3 \leq t \leq n$, and back from v_{k-2}^n to v_{k-2}^3 , and repeated until v_{j+1}^3 , which then go through subpath of P_3 backwards to v_u^3 and go to corresponding vertex in G_4 , then go through the subpath of P_4 to v_j^4 , then jump to the corresponding vertex in G_5 and repeat until v_j^n in G_n , which then go through all the v^t for $n \geq t \geq 2$, then go through the subpath of P_2 backwards to v_{u-1}^2 , and go through all the v_{u-1}^t for $3 \leq t \leq n$ and back from v_{u-2}^n to v_{u-2}^2 , and repeat until v_1^2 which then end with v_1^1 and form a Hamiltonian cycle of

$G \times C_n$ as shown in Figure 1.

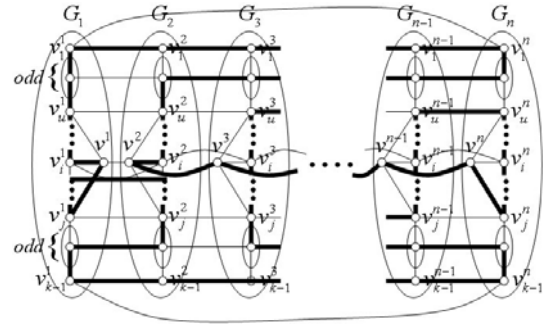


Figure 1. A Hamiltonian cycle in Case 1.1 of $G \times C_n$.

Case 1.2. n is odd but both $u-1$ and $k-1-j$ are even. Similar to Case 1.1, rather than go through v^t from v_j^n , we will have to use v_u^n since with odd n the subpath P_n will go backwards from v_j^n to v_u^n . The Hamiltonian cycle of $G \times C_n$ in this case is shown in Figure 2.

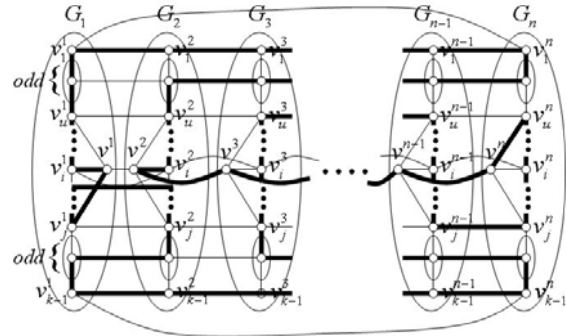


Figure 2. A Hamiltonian cycle in Case 1.2 of $G \times C_n$.

Case 2.1. n is even, but both $u-1$ and $k-1-j$ are odd. In this case, when going back and forth between G_1 and G_2 , because $k-1-j$ is odd, it will end with v_{k-1}^1 instead of v_{k-1}^2 , so it will have to go through all v_{k-1}^t from $t=n$ back to $t=3$ then repeat the same routing as in Case 1.1. Because of $u-1$ is odd, the routing will end with v_1^n instead of v_1^2 which then back to v_1^1 to form a Hamiltonian cycle. The Hamiltonian cycle of $G \times C_n$ in this case is shown in Figure 3.

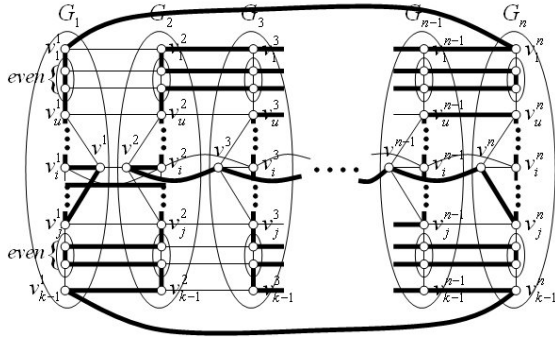


Figure 3. A Hamiltonian cycle in Case 2.1 of $G \times C_n$.

Case 2.2. n , $u-1$ and $k-1-j$ are all odd. This case may be obtained by combining Case 2.1 (for the routing from v_1^1 to v_j^3) with Case 1.2 (for the routing from v_j^3 to v_1^1), and the Hamiltonian cycle of $G \times C_n$ in this case is shown in Figure 4.

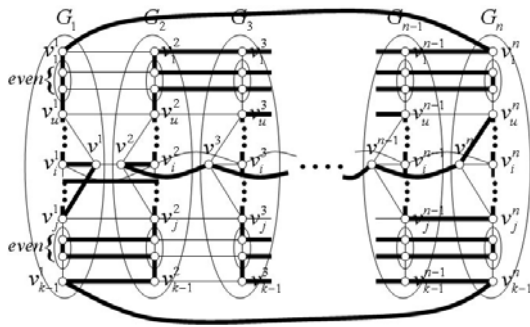


Figure 4. A Hamiltonian cycle in Case 2.2 of $G \times C_n$.

Case 3.1. One of $u-1$ and $k-1-j$ is odd (say $u-1$ is odd), and n is even. In this case, the routing will be the same as Case 1.1 except that at the end of the cycle, v_1^1 is adjacent from v_1^n instead of v_1^2 . The Hamiltonian cycle in this case is shown in Figure 5.

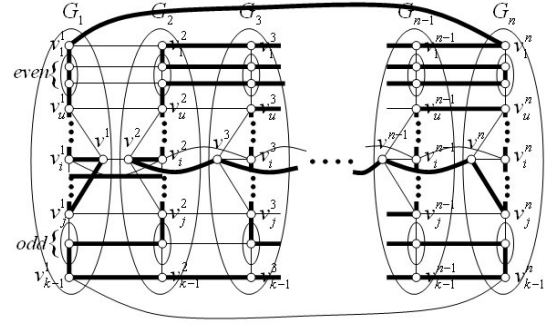


Figure 5. A Hamiltonian cycle in Case 3.1 of $G \times C_n$.

Case 3.2. One of $u-1$ and $k-1-j$ is odd (say $u-1$ is odd), and n is odd. In this case, the routing will be the same as Case 1.2 except that at the end of the cycle, v_1^1 is adjacent from v_1^n instead of v_1^2 . The Hamiltonian cycle in this case is shown in Figure 6.

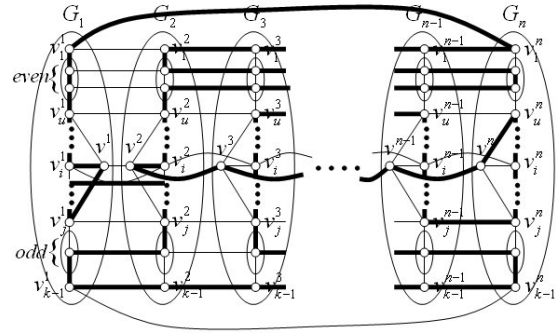


Figure 6. A Hamiltonian cycle in Case 3.2 of $G \times C_n$.

After considering these only six cases of the graph $G \times C_n$, we prove that $G \times C_n$ admits a Hamiltonian cycle. ■

Lemma 2 Suppose the graph G is a hypotraceable graph, then $G \times Q_1$ admits a Hamiltonian cycle.

Proof. $G \times Q_1$ can be represented as a perfect matching between the corresponding vertices of two identical graphs G_1 and G_2 . Since G_1 and G_2 are two identical hypotraceable graphs, let $P_1: v_1^1, v_2^1, \dots, v_{k-1}^1$ and $P_2: v_1^2, v_2^2, \dots, v_{k-1}^2$ be two

identical Hamiltonian paths of $G_1 - v^1, G_2 - v^2$ respectively. Suppose v_u^t, v_i^t, v_j^t for $u < i < j$ be three vertices adjacent to v^t in G_t for $1 \leq t \leq 2$, consider following two cases:

Case 1. For either $i - u$ or $j - i$ is even (say $i - u$ is even). Consider a cycle start from v_1^1 , go through subpath of P_1 to v_{u+1}^1 , then go back and forth between corresponding vertices of G_1 and G_2 until v_{i-1}^2 , then go through subpath of P_2 to v_{k-1}^2 then back to v_{k-1}^1 and go backwards in the subpath of P_1 to v_i^1 , then pass v^1 and v^2 to v_u^2 and go backwards in the subpath of P_2 to v_1^2 then back to v_1^1 , which is a Hamiltonian cycle as shown in Figure 7.

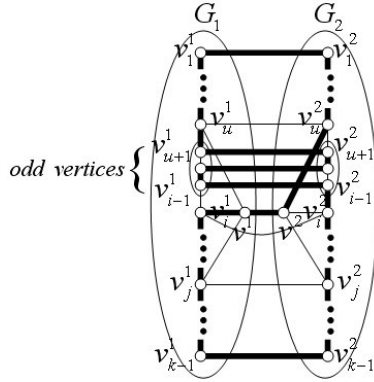


Figure 7. A Hamiltonian cycle in Case 1 of $G \times Q_1$.

Case 2. For both $i - u$ and $j - i$ are odd. Then consider the cycle similar to Case 1 but go back and forth between G_1 and G_2 from v_{u+1}^1 to v_{i-1}^1 , then go to v_i^1 and v_i^2 , then go back and forth between G_1 and G_2 from v_{i+1}^2 to v_{j-1}^2 then follow the subpath of P_2 to v_{k-1}^2 then v_{k-1}^1 and go backward in the subpath of P_1 to v_j^1 , then pass v^1 and v^2 to v_u^2 , and go backwards in P_2 to v_1^2 then back to v_1^1 , which is a Hamiltonian cycle as shown in Figure 8.

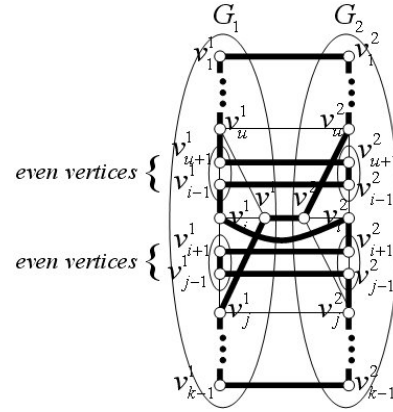


Figure 8. A Hamiltonian cycle in Case 2 of $G \times Q_1$.

Therefore, by Case 1 and Case 2, $G \times Q_1$ admits a Hamiltonian cycle. ■

Since C_n , $n \in 2^k$ is a subgraph of a hypercube graph, Theorem 2 comes directly from Theorem 1 and Lemma 2.

Theorem 2 Let G be a hypotraceable graph, then $G \times Q_n$ admits a Hamiltonian cycle.

Definitely, C_n is also a subgraph of a Hamiltonian graph of order n . Hence the following corollary may be implied by Theorem 1.

Corollary 1 Let G_1 be a hypotraceable graph and G_2 be a Hamiltonian graph. Then $G_1 \times G_2$ admits a Hamiltonian cycle.

3 Conclusion

In this paper, we propose several construction schemes for the Cartesian product of a hypotraceable graph with a Hamiltonian graph. Applying these schemes, we can construct a series of new Hamiltonian graphs.

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